

Study tools for students

What Turbo AI does, what the research says works, and how Smart Learning can do it better for Jamaican students

Research and blueprint. Prepared 10 October 2026. This belongs to the Learning product only (assessments, results and report cards are not touched).

1. The short version

What Turbo AI is. A consumer study app, formerly called TurboLearn. A student uploads a PDF, a YouTube link or a lecture recording and Turbo produces notes, flashcards, multiple choice quizzes with explanations, a podcast-style audio summary, and a chat box that answers questions about those notes. It runs on web, iOS and Android, has a free tier and a paid tier of roughly \$10 to \$20 a month, and its terms say users must be 16 or older.

What we found.

- **The idea is sound, the product is thin.** Turbo is a clean pipeline from "my material" to practice. Its "AI tutor" is a chatbot over your own notes. It is not a structured tutor.
- **The research favours the quiz and flashcard half, not the notes half.** Practice testing and spaced practice are the two best-supported study methods; summarising and rereading are among the weakest. Generated notes are the least valuable thing Turbo makes.
- **Unguarded AI help can lower exam results.** In a randomised trial of nearly a thousand high school maths students, those given a plain ChatGPT-style helper did better in practice and worse on the later exam than students with no AI at all. A tutor built to give hints removed most of that harm.
- **Smart Learning already has most of the hard parts.** A hint-first AI tutor with safety flags and school consent, spaced-repetition flashcards that work offline, three-level lesson checks, topic tracking and a library. What is missing is the front door: letting a student (or teacher) turn their own material into practice.

What we would build, in order.

1. **Teacher study packs:** a teacher uploads or picks material and Smart Learning drafts flashcards and a quiz with explanations; the teacher reviews and assigns it to a class. Safest to start with, because the content is teacher-owned and checked.
2. **Student study packs:** a student uploads their own notes (a PDF, a document, or a photo of handwritten notes) and gets flashcards and a quiz that cite the page they came from.
3. **Ask your material:** the existing tutor, extended to answer questions from the student's own pack with citations, hint-first.
4. **Today's practice:** one daily set that mixes due flashcards, weak topics from My Topics and new quiz questions, so spaced practice happens by default.

What we would leave out. Live lecture recording (consent and privacy), "brainrot" video gimmicks, public note sharing, cross-app tracking, and scraping YouTube.

Cost. On Anthropic's published prices, a study pack from a 20-page document costs about 9 cents with Claude Sonnet 5.5 or under half a cent with Claude Haiku 5.5. A student who makes 8 packs and

sends 150 chat messages a month costs about \$1.60 on Sonnet or about \$0.10 on Haiku. These are estimates to be measured in the pilot (section 9).

Decisions needed from you are in section 12.

2. What Turbo AI is and what it actually does

Everything below is taken from Turbo's own site, its blog and terms of service, and a small number of third-party reviews. Where the only source is Turbo itself, that is stated. Turbo's marketing numbers (millions of users, "3x faster", "99% accuracy") could not be verified and are not used in this blueprint.

Feature	What Turbo says it does	Evidence	Worth taking?
Notes from uploads	PDFs, YouTube links, audio and live recordings become organised notes in a Google Docs-style editor, in nested folders that sync across web, phone and tablet.	Turbo's site and blog	Partly. Useful as a reading aid and as the base for practice, not as a study method on its own.
Flashcards	Generated automatically for key concepts; "smart spaced repetition adapts to your pace"; Learn Mode sorts cards into New, Learning and Mastered.	Turbo's site and blog	Yes. We already have the spacing engine; we need the generation.
Quizzes	Multiple choice from your notes; choose topics or chapters; adjustable difficulty from basic to exam level; an explanation for every answer.	Turbo's site	Yes. The most valuable feature.
Chat with your notes	A chatbot inside the notes that answers questions without pasting anything and "answers at the level of the material".	Turbo's blog; no detail on citations	Yes, with citations and hint-first behaviour.
Podcast audio	Notes turned into a quick summary or deep dive audio, downloadable for offline listening.	Turbo's site	Later. Helpful on long bus rides; needs a speech service.
Lecture recording and notetaker	Records a live lecture, transcribes it and makes notes.	Turbo's site; reviewers report failed audio uploads	No. Consent, privacy and classroom policy problems outweigh the value.
Book library	Browse thousands of academic books, read, highlight and generate notes and flashcards from any chapter.	Turbo's site	Yes, later, through the Smart Learning Library once it has books.
Collaboration	Share notes with classmates, live editing, inline comments.	Turbo's site	Teacher-controlled class sharing only. No open sharing of student uploads.
STEM formatting	Keeps maths formulas, chemical equations and code readable.	A review; Turbo's own blog admits its flashcards "didn't always include equations"	Essential for maths and science. We already render maths.

Feature	What Turbo says it does	Evidence	Worth taking?
Premade study guides	A library of AP study guides.	Turbo's site	Our equivalent is curriculum-aligned lessons and packs built on the National Standards Curriculum.

Pricing, as far as it can be established

Turbo's site gives no prices. Third-party listings disagree: a free tier with tight monthly limits (one listing says about 2 hours of lecture processing, 1 PDF, 5 quiz sessions and 10 chat messages), and a paid tier between \$5.99 and \$49 a month, most often quoted at \$9.99 a month billed yearly or \$19.99 monthly. Treat all of these as rough.

What its terms and reviews tell us

- Minimum age 16; 16 and 17 year olds need a parent or guardian's consent. Our students are mostly 11 to 18, so Turbo itself would not be suitable for most of them without a school arrangement.
- Uploads are processed by third-party AI providers it names: Grok, Claude, OpenAI, Anthropic, Together AI, Google and Modal Labs. It may change them at any time.
- It says it is "not the system of record" for your files and makes no promise to keep inputs. It does not clearly say whether your content trains AI.
- Users are "solely responsible" for having the right to upload, and are told not to upload health, financial or ID information.
- Outputs may contain errors and "hallucinations" and users must review them. A third-party review found summaries sometimes omit details; another reports bugs, slow support and users feeling pressed to subscribe to keep their notes.
- The same review says users are tracked across other apps and sites by user and device ID. That is not acceptable for a school product.

What could not be verified

- User numbers, "3x faster" learning and any accuracy percentage: no method or source given.
- Exact free and paid limits, supported languages, and how the spaced repetition schedule actually works.
- Whether its "AI tutor" does anything beyond answering questions about your notes. Nothing found suggests it teaches in steps or adapts a learning path.

3. What the research says works

Before choosing features it is worth knowing which study behaviours actually help, and where AI helps or harms.

Finding	What it showed	What it means for us
Practice testing and distributed practice are the best-supported study techniques	A review of ten common techniques (Dunlosky and colleagues, 2013) rated practice testing and spacing as high utility across ages. Summarising, highlighting and rereading were rated low, although students use them most.	Put quizzes, flashcards and spacing at the centre. Treat generated notes as the starting point for practice, not the product.

Finding	What it showed	What it means for us
Unrestricted AI help can hurt later performance	In a randomised trial with nearly 1,000 high school maths students (Bastani and colleagues, PNAS 2025), a plain GPT-4 helper improved practice scores but students scored 17% lower on the later exam than students who never had it. A tutor built to give hints largely removed the harm.	The tutor must give hints before answers and ask the student to try first. Our existing tutor already does this; every new feature must keep it.
Students misjudge their own learning	In the same study students with the unrestricted helper believed they had learned as much as their peers while doing worse on the exam.	Show students what they can actually do (quiz results, topics that need work), not how confident they feel.

Practical rules that follow: default to testing rather than summarising; make the student attempt before they see an answer; show them their own gaps; and make answers traceable to the source so wrong ones can be caught.

4. Where Smart Learning is today

Capability	Smart Learning now	Turbo AI	Gap
Spaced repetition flashcards	Yes. Leitner boxes (back after 0, 1, 3, 7 and 21 days), study sessions, offline use. Students make their own decks.	Yes	We need AI to make the cards
AI tutor	Yes. Asks about one teacher lesson, hints before answers, safety flags that an adult reviews, school consent, rate limits, blocked during exams.	Chat over notes	Ours is safer; it is limited to teacher lessons
Practice questions	Yes. Lesson checks at Support, Core and Stretch levels, AI-drafted by teachers.	Yes	Students cannot make their own
Topic and progress tracking	Yes. My Topics (from marked results), My Progress, catch-up for absent students.	Weak areas	None for tracking; needs practice results added
Library and audio	Library built, catalogue empty; read-aloud button on content.	Book library, podcasts	Needs books; audio summaries missing
Upload own material	No	Yes	Main gap
Generated flashcards and quiz from material	No	Yes	Main gap
Answers that cite the source	Lesson-bound, no page citations	Not described	Add
Photo of handwritten notes	No	Not described	Add (strong fit for our students)

What we have that Turbo does not

The National Standards Curriculum, 5E and depth-of-knowledge alignment; teachers in the loop; school consent and oversight; flagged conversations for adults; an exam lock; offline and low-data use; and one login with Smart Assess and the school's topic list.

5. Principles the build must follow

1. **Testing before telling.** The default output is questions and flashcards, not summaries.
2. **Always grounded.** Every card, question and answer points to the page or passage it came from. Nothing is shown without a source.
3. **Hints before answers.** The student attempts first; the tutor guides. This is already the tutor's rule and is non-negotiable.
4. **A teacher can see and shape it.** Teachers can create packs for their class, review what students are told, and see flagged conversations. Students' private packs are not read by teachers without a flag.
5. **Works on a poor connection.** Generated sets can be used offline once downloaded; generation itself is a short online step with clear progress.
6. **Cost is capped per student.** Monthly allowances like the ones already used for AI drafting, so a class cannot run up an unexpected bill.
7. **Never in an exam.** Study tools and the tutor are unavailable while a student has an exam or test in progress.
8. **Learning only.** Practice results stay in Smart Learning. They never become marks, report card grades or Assess results.

6. The features

Size key: S is 1 to 2 days of build, M is 3 to 5 days, L is 1 to 2 weeks. Review, testing and a classroom pilot take longer than the build itself.

F1. Teacher study packs (build first)

What the teacher does	Opens "New study pack", uploads a PDF or Word file (or picks one of their own lessons), chooses subject, grade and a topic from the school topic list, and presses Draft. In about a minute there are 15 to 25 flashcards and 10 quiz questions at Support, Core and Stretch levels, each with an explanation and the page it came from. The teacher edits, removes anything wrong, approves, and assigns the pack to a class.
What the student sees	A pack in their lessons list. "Study" opens flashcards with spaced repetition. "Quiz" asks a question first and explains afterwards. Wrong answers come back later.
Why first	The content is teacher-owned and checked before students see it, so there is no upload risk and no unreviewed AI content. It also gives us real accuracy and cost numbers before students can upload anything.
Reuses	The flashcard decks and spacing engine, lesson checks and their three levels, the teacher review pattern used for AI-drafted lesson checks, the monthly AI allowance, school topics, offline use.
New	Study pack tables, a draft endpoint, a review and assign screen.
Size	L

F2. Student study packs

What the student does	"Make a study pack": uploads their own notes as a PDF, a Word file or photos, picks the subject and gets flashcards and a quiz. Every card and question shows the page or photo it came from.
Rules	Up to 30 pages or 10 MB per upload; 10 packs a month (adjustable); for the student's own studying and never shared with other students; a warning not to upload personal details and an automatic check for phone numbers and ID numbers; the original file is deleted within 24 hours; the school can switch the feature off.
Reuses	The whole F1 pipeline, the school consent switch used for the AI tutor, the existing rate limits and flag review.
New	Student upload screen, personal-data check, per-student allowance, "report a wrong card" button.
Size	M after F1
Main risks	Copyright of what is uploaded, personal data in uploads, and misuse. See section 8.

F3. Ask your material

What the student does	Asks a question inside a pack. The answer is short, quotes the page it came from, and for practice questions gives a hint first. If the answer is not in their material the tutor says so and offers a clearly marked general explanation.
How	The existing tutor, with the lesson replaced by the pack's text. Small packs are sent whole with caching, which keeps messages cheap. Large packs retrieve the most relevant passages.
Reuses	The tutor rules, safety flags, school consent, rate limits and the exam block.
Size	M

F4. Today's practice

What the student sees	One card on the Smart Learning home page: "12 cards due, 4 questions on Fractions, 3 new". A ten minute session mixes due flashcards, topics that My Topics shows are weak, and new quiz questions, so spaced practice happens by default instead of by willpower.
How	Uses spacing dates already stored, weak topics already computed, and unseen questions. No new AI cost.
Important	Practice results are recorded separately from marks and never feed report cards or Assess.
Size	M

F5. Photo of notes

What the student does	Photographs pages of handwritten notes or the board. The app reads the text and shows it back so the student can correct it before anything is generated.
Why it fits us	Most of our students' notes are handwritten. Turbo AI does not describe this at all.
Risks	Poor photos and handwriting, diagrams and equations, and people in photos. Mitigated by the correction step, a "no faces" prompt and deleting the image after reading.
Size	M

F6. Audio review

What the student does	Presses "Listen" on a pack and hears the key points read aloud.
How	Start with the speech built into the phone or browser (free, works offline, quality varies). Move to a paid speech service only if students use it and quality complaints justify the cost.
Size	S to M

F7. Library chapter to pack

What it is	Once the Smart Learning Library has books, a student or teacher can turn a chapter into a pack with the same pipeline.
Size	S after F1

What we will not build

Turbo feature	Why we leave it out
Live lecture recording and AI notetaker	Recording a classroom raises consent, privacy and school policy problems, and the learning value is lower than practice. Turbo's own terms push the responsibility onto students.
"Brainrot" and celebrity video explainers	Promotional gimmicks with no evidence of learning benefit.
Open sharing of notes	Student uploads can contain personal and copyrighted material. Sharing stays teacher-controlled and class-only.
YouTube link ingestion	Unreliable transcripts and platform terms. Teachers already add video links to the Videos page.
Cross-app tracking and a paywall	Not appropriate for a school product.

7. How it is built

One pipeline, used by every feature

1. **Upload and check.** File type, size and page count; a scan for obvious personal data; the student's allowance.
2. **Read the text.** Use the PDF's own text where there is some (cheap and exact). Scanned pages and photos go through the vision model. Word files are converted. Page numbers are kept.
3. **Split into passages** with numbers and page references.
4. **Draft.** Ask the model for structured output: cards (front, back, passage) and questions (stem, options, answer, explanation, passage, level). The instruction is to use only the material and write maths in plain notation.
5. **Check.** The structure is valid; every cited passage exists; the key words of the answer appear in the cited passage, otherwise the item is dropped or marked "check this"; duplicates are removed.
6. **Save.** Cards go into the existing flashcard decks. Questions go into a new table. The pack stores where each item came from, not the file.
7. **Study.** Spacing, quiz attempts and a wrong-answer review.

8. **Clean up.** The original is deleted within 24 hours. Deleting a pack deletes its passages, cards and questions.

What is added to the database (proposed)

Table	Purpose
learning_study_packs	One row per pack: owner, teacher or student kind, title, subject, grade, topic, status, class assignment, approval.
learning_study_passages	The extracted text split into numbered passages with page numbers. This is what answers are checked against.
learning_study_questions	Quiz questions with level, options, answer, explanation, source passages, hidden flag.
learning_study_attempts	A student's answers, used for practice and weak topics only. Never used for marks.
learning_study_reports	A student or teacher marking a card or question as wrong, with a reason.
learning_study_usage	Monthly counts and token use per student, for allowances and cost tracking.
Flashcard decks and cards (existing)	Gain a pack reference and source passages. The spacing engine and offline support are unchanged.
Tutor conversations (existing)	Gain an optional pack reference. Flagging, consent and review are unchanged.

Rules in the database

- A student sees only their own packs and class packs a teacher approved for their class.
- A teacher sees the packs they made and class packs they assigned. Teachers do not browse students' private packs; flagged conversations and reported items reach the teacher or head of department as they do today.
- Only the owner can delete a pack, and deleting it removes everything derived from it.
- The school leadership switch that already governs the AI tutor also governs study packs. Nothing works until it is on.
- Rules like these are tested the way the rest of the project is: sandbox tests for every rule and a probe from a student account that tries to read, change and delete what it should not.

Models and limits

- Start with Claude Sonnet 5.5 for quality during the pilot; measure Claude Haiku 5.5 on the same 100 items. If accuracy is within tolerance, use Haiku to draft cards and questions and keep Sonnet for the tutor.
- Drafting runs in two steps (cards, then questions) with visible progress, so each call stays inside the 60 second limit of the hosting.
- Per student: 10 packs and 100 tutor messages a month by default, plus the burst limits already in place. All adjustable by the school.
- Failures use the same plain-language messages and alerts as the existing AI features, and never leave a half-made pack on screen.

8. Safety, privacy and rules

Concern	Rule
Age and consent	Turbo is for 16 and over. Ours is run through the school, so the school leadership approves the feature (as with the AI tutor) and informs parents. Students are mostly 11 to 18.
Personal data in uploads	A clear warning, an automatic check for phone numbers, ID numbers and addresses, and the original file deleted within 24 hours. Prompts contain only the material.
Copyright	Uploads are for the student's own studying, never shared with other students, deleted after reading. Teacher packs are made from materials the school may use. The school should say in writing what students may upload.
Retention	Originals 24 hours. Packs until the student deletes them, with automatic clearing after 12 months unused (to be confirmed).
Who processes the data	Anthropic, through our existing AI connection. It must be named in the school's privacy notice. Confirm in Anthropic's current commercial terms that API data is not used for training. No other processor is added.
Accuracy	Every item is grounded in a passage and checked against it; students can report a wrong card; teachers review class packs; the pilot includes a teacher accuracy audit.
Over-reliance	Attempt first, hints before answers, and a tutor that does not do the work. Students are shown their own results, not just their confidence.
Exams and integrity	Study tools and the tutor are blocked while an exam or test is in progress. Live exam questions from Assess are never read into a pack.
Wellbeing and content	The existing flagging sends worrying messages to an adult; the tutor already declines unsuitable topics.
Cost abuse	Per student allowances, burst limits and a monthly cost report for the school admin.
Jamaican law	The Data Protection Act applies to children's personal data. Before the pilot, write a short data protection impact assessment and have the school's legal advisor confirm the wording of the notice and the consent letter.

9. What it costs to run

These use Anthropic's published prices (Sonnet 5.5: \$2 per million input tokens and \$10 per million output; Haiku 5.5: \$0.10 and \$0.50; cached reads far cheaper; a 50% discount for non-urgent batch work). The token counts are assumptions to be measured in the pilot, not measurements.

Operation	Assumed tokens	Sonnet 5.5	Haiku 5.5
Study pack from a 20 page document	15,000 in, 6,000 out	about \$0.09	about \$0.005
One tutor message on a cached pack	15,000 cached, 600 new, 300 out	about \$0.006	about \$0.0004
One student for a month (8 packs, 150 messages)		about \$1.60	about \$0.10
Whole school, 30% of 1,800 students active		about \$875 a month	about \$50 a month

- The allowance caps one student at roughly the Sonnet figure above, so the worst case is predictable.

- PDFs full of images cost many more tokens than text, so the pipeline reads the text layer first and only uses the vision model where needed.
- Caching and batch processing can reduce these further. The pilot will give real numbers before anything is switched on school-wide.

10. Build plan

Phase	What	Size	Done when
0. Decide and prepare	Your decisions (section 12), the policy wording, allowances, a short impact assessment, choice of pilot classes.	S	The decisions are made and the school leadership has approved the wording.
1. Teacher study packs	F1: upload or pick material, draft cards and quiz, review, assign; students study offline-capable.	L	A teacher can make and assign a pack; the accuracy audit passes; cost per pack is known.
Pilot A	Two classes, four weeks, teacher packs only.		Measures below are met.
2. Student study packs and Ask your material	F2 and F3 with personal-data check, allowances and the report button.	L	Students can make packs and ask questions; safety tests pass.
3. Today's practice and photo of notes	F4 and F5.	M each	Daily practice mixes spacing and weak topics; photos read correctly after correction.
4. Audio and library	F6 and F7.	S to M	Listening works offline; a library chapter becomes a pack.

How we will know it works

- Weekly active students and cards reviewed per student.
- Share of cards reaching box 4 or higher after four weeks (spacing is working).
- Quiz accuracy on the same topic two weeks later, and results on the teacher's next test against a similar class without packs.
- Teacher-flagged errors: fewer than 5 per 100 items and no serious ones.
- Tutor flags and how they were handled.
- Cost per active student against the estimate.
- A short student and teacher survey added to the existing survey.

Tests every phase must pass

- Teachers check 100 generated items: at least 95 correct and correctly cited.
- Prompts that try to get exam answers, personal information or harmful content are declined or flagged.
- A student with an exam in progress cannot open the tools.
- A student cannot read, change or delete another student's pack (sandbox tests and a live probe).
- Study works offline once a pack is downloaded.
- No uploaded original remains after 24 hours.

11. Risks

Risk	How it is handled
Wrong or misleading cards	Grounding and checking, a report button, teacher review of class packs, and an accuracy audit before wider use.
Students lean on the tutor and learn less	Attempt first, hints before answers, practice results shown to the student, and the exam block.
A data protection complaint	School approval and parent notice, minimal retention, no file kept, a named processor and a written impact assessment.
A copyright complaint	Uploads are private, never shared, deleted after reading; class packs only from material the school may use.
Cost surprise	Allowances, burst limits, a monthly cost report and a cheaper model fallback.
Weak results on maths, diagrams or handwriting	Correction step for photos, plain notation for maths, and a pilot that tests these subjects first.
Teacher review becomes a chore	A fast review screen, bulk approve with spot checks, and sensible defaults.
Scope creep into Assess	The product boundary rule: practice results never become marks or report card entries.
Dependence on one AI supplier	All AI calls already go through one wrapper, so a second supplier can be added without changing the features.

12. Decisions needed from you

	Decision	My recommendation
1	Approve building teacher study packs first?	Yes. It is the safest way to get real accuracy and cost numbers.
2	Can students upload their own material in the pilot, or teacher-shared only at first?	Teacher packs for Pilot A, student upload in Pilot B for two classes.
3	Which file types first?	PDF and photos. Word files after.
4	Monthly allowance per student?	10 packs and 100 tutor messages, adjustable by the school.
5	How long to keep packs and passages?	Originals 24 hours; packs until deleted; clear after 12 months unused.
6	Who approves the feature for the school?	The same leadership switch as the AI tutor.
7	Which AI model to start with?	Sonnet 5.5, with Haiku 5.5 tested on 100 items.
8	Pilot classes?	The Grade 9 mathematics class used for the demo data, plus one English class.
9	Audio?	Phone or browser speech first.
10	Name in the product?	"Study packs", under Smart Learning.

13. Sources

Turbo AI (vendor material).

- [1] Turbo AI home: <https://www.turbo.ai/>
- [2] Turbo AI for students: <https://www.turbo.ai/for-students>
- [3] Turbo AI terms of service: <https://www.turbo.ai/terms-of-service>
- [4] Turbo AI vs ChatGPT (Turbo's own blog): <https://www.turbo.ai/blog/turbo-vs-chatgpt>
- [5] Turbo AI vs StudyFetch (Turbo's own blog): <https://www.turbo.ai/blog/studyfetch-vs-turbo>

Reviews.

- [6] HyScaler review (marketing in tone; pricing and tracking notes): <https://hyscaler.com/insights/discover-turbolearn-ai/>
- [7] Tools for Humans listing: <https://www.toolsforhumans.ai/ai-tools/turbolearn-ai>

Research.

- [8] Dunlosky and colleagues (2013), Improving students' learning with effective learning techniques: <https://doi.org/10.1177/1529100612453266>
- [9] Bastani and colleagues (2025), Generative AI without guardrails can harm learning, PNAS: <https://doi.org/10.1073/pnas.2422633122>
- [10] Wharton summary of the Bastani study: <https://knowledge.wharton.upenn.edu/article/without-guardrails-generative-ai-can-harm-education>

Prices.

- [11] Anthropic pricing: <https://platform.claude.com/docs/en/about-claude/pricing>

Limits of this research. Turbo AI is a closed product: how its generation and spacing really work, its accuracy and its limits could not be tested directly without an account. Third-party reviews were of mixed quality and are flagged where used. Vendor claims are labelled as claims.